

Hi Candidate,

As part of the interview process, we would like you to complete a system design assignment focused on designing and building components of an Internal AI Knowledge Platform.

The goal of this assignment is to evaluate your understanding of:

- API Design
 - RAG / Semantic Search Systems
 - AI Infrastructure
 - Database Design
 - Scalability & Reliability
 - Production Engineering Practices
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Assignment Overview

Design a backend platform that allows internal developers to:

1. Upload documents/code files
2. Query documents using natural language
3. Delete documents
4. Perform semantic search over uploaded knowledge
5. Access LLMs through a centralized AI Gateway

The platform will be used internally by approximately 100 developers.

Requirements

Part 1 — Document APIs

Design APIs for the following:

1. Upload Document API

Requirements:

- Accept PDF, markdown, text documents, or code files

- Extract document content
- Chunk documents
- Generate embeddings
- Store metadata
- Support asynchronous processing

Example:

POST /documents

2. Query Document API

Requirements:

- Accept natural language query
- Retrieve semantically relevant chunks
- Support metadata filtering
- Return ranked results

Example:

POST /query

3. Delete Document API

Requirements:

- Delete document metadata
- Delete embeddings/chunks
- Handle partial failures gracefully
- Explain soft delete vs hard delete approach

Example:

DELETE /documents/{id}

Part 2 — Semantic Search Service

Design a semantic retrieval system that:

- Generates embeddings for search queries
- Performs vector similarity search
- Retrieves top-K relevant chunks
- Supports metadata filtering
- Handles large-scale document retrieval efficiently

Please explain:

- Chunking strategy
 - Embedding lifecycle
 - Similarity search approach
 - Vector database choice
 - Ranking/reranking strategy
 - Failure handling
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Part 3 — Database Design

Design schemas/storage strategy for:

- Documents
- Chunks
- Embeddings
- Query Logs

Please explain:

- Indexing strategy
 - Partitioning
 - Metadata modeling
 - Query patterns
 - Caching
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Deliverables

Please submit:

1. System architecture diagram
2. API design/specification

3. Database schema design
 4. Brief explanation of scaling strategy
 5. Any assumptions/tradeoffs made
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Evaluation Criteria

We will evaluate:

- System design clarity
 - Production readiness
 - API quality
 - AI/RAG understanding
 - Database design quality
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Time Expectation

Expected effort: 1Day

Upload your submissions on github and share the repo link or share the folder with relevant documents and code implementations

You are free to use diagrams, markdown, PDFs, or presentation slides.

Looking forward to reviewing your solution.

Best Regards,
Hiring Team